

CSE 717 — Information Security

Topic 7: Number Theory & Modular Arithmetic

Complete Past Paper Solutions: 2020–2024

Source: Lc#5 (GCD, Extended Euclidean, linear congruence), Lc#6 (Fermat, Euler) — CRT, Miller-Rabin, discrete log/cyclic groups are no-source cheat sheet

HIGHEST-YIELD TOPIC: 5/5 years, 5–9 marks. Every Topic-7 question 2020–2024 included. Zero omissions.

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Quick Reference — 8 Recurring Sub-Patterns (CHEAT SHEET)

CONFIRMED EVERY YEAR — foundation for RSA (Topic 8) and Digital Signatures

Recurring numeric “seeds”: $3^{20x} \bmod 11$ and $7^{1000} \bmod (10 \text{ or } 11)$ (Fermat/Euler combo, appears 2020/2021/2023/2024). Extended Euclidean for multiplicative inverse appears almost every year. CRT appears 2020 & 2023. Miller-Rabin + discrete log appear 2020–2022. **Patterns A–E sourced from Lc#5/Lc#6; patterns F–H are NOT in any 2026 Lc material — pure cheat sheet, memorize the algorithm + 1 worked example each.**

A. GCD via the Euclidean Algorithm

Repeatedly apply $\gcd(a, b) = \gcd(b, a \bmod b)$ until the remainder is 0. The last nonzero remainder is the GCD.

Worked example (Lc#5): $\gcd(414, 662)$

$$662 = 1(414) + 248, \quad 414 = 1(248) + 166, \quad 248 = 1(166) + 82, \quad 166 = 2(82) + 2, \quad 82 = 41(2) + 0$$

$$\Rightarrow \gcd(414, 662) = 2.$$

Worked example (Lc#5): $\gcd(252, 198)$

$$252 = 1(198) + 54, \quad 198 = 3(54) + 36, \quad 54 = 1(36) + 18, \quad 36 = 2(18) + 0$$

$$\Rightarrow \gcd(252, 198) = 18.$$

B. Extended Euclidean Algorithm — GCD as a Linear Combination & Multiplicative Inverse

Back-substitute through the Euclidean steps to write $\gcd(a, b) = sa + tb$ for integers s, t . If $\gcd(a, b) = 1$, then $s \equiv a^{-1} \pmod{b}$ (or $t \equiv b^{-1} \pmod{a}$).

Worked example (Lc#5): $18 = \gcd(252, 198)$ as a linear combination

$$18 = 54 - 1(36) = 54 - 1(198 - 3(54)) = 4(54) - 198 = 4(252 - 198) - 198 = 4(252) + (-5)(198)$$

$$\Rightarrow 18 = 4(252) + (-5)(198)$$

Worked example (2021 A3): multiplicative inverse of 1234 mod 4321 Euclidean steps: $4321 = 3(1234) + 619$, $1234 = 1(619) + 615$, $619 = 1(615) + 4$, $615 = 153(4) + 3$, $4 = 1(3) + 1$. Back-substitute: $1 = 309(4321) - 1082(1234)$

$$\Rightarrow 1234^{-1} \equiv -1082 \equiv 4321 - 1082 = \boxed{3239} \pmod{4321}$$

(Check: $1234 \times 3239 = 3,996,926 = 925 \times 4321 + 1$.)

Worked example (2020 Q3): multiplicative inverse of 550 mod 1769 Euclidean steps reduce to $1 = 550(550) - 171(1769)$

$$\Rightarrow 550^{-1} \equiv \boxed{550} \pmod{1769}$$

(Check: $550 \times 550 = 302,500 = 171 \times 1769 + 1$. Pleasant surprise — 550 is its own inverse!)

C. Linear Congruence $ax \equiv b \pmod{m}$

Steps:

1. Check $\gcd(a, m) = 1$ (otherwise need $\gcd(a, m) \mid b$ for solutions to exist).
2. Find $a^{-1} \pmod{m}$ via Extended Euclidean (pattern B).
3. Solution: $x \equiv a^{-1} \cdot b \pmod{m}$. General solution: $x_0 + mk$ for integer k .

Worked example (Lc#5): solve $33x \equiv 38 \pmod{280}$ $\gcd(33, 280) = 1$. Extended Euclidean: $1 = 17(33) - 2(280) \Rightarrow 33^{-1} \equiv 17 \pmod{280}$.

$$x \equiv 17 \times 38 = 646 \equiv 646 - 2(280) = \boxed{86} \pmod{280}, \quad \text{general: } 86 + 280k$$

Worked example (2023 Section B Q7(b)): solve $33x \equiv 18 \pmod{280}$ Same $33^{-1} = 17$.

$$x \equiv 17 \times 18 = 306 \equiv 306 - 280 = \boxed{26} \pmod{280}, \quad \text{general: } 26 + 280k$$

D. Fermat's Little Theorem

$$a^{p-1} \equiv 1 \pmod{p}, \quad \text{where } p \text{ is prime and } \gcd(a, p) = 1.$$

Steps to compute $a^n \pmod{p}$:

1. Write $n = (p-1) \cdot k + r$, i.e. $r = n \pmod{p-1}$.
2. Then $a^n = (a^{p-1})^k \cdot a^r \equiv 1^k \cdot a^r \equiv a^r \pmod{p}$.
3. Compute $a^r \pmod{p}$ directly.

Worked example (Lc#6): $5^{301} \pmod{11}$ $5^{10} \equiv 1 \pmod{11}$. $301 = 10(30) + 1 \Rightarrow 5^{301} \equiv 5^1 = \boxed{5} \pmod{11}$.

Quick practice (Lc#6): $3^{31} \pmod{7}$, $2^{35} \pmod{7}$, $128^{129} \pmod{17}$ $3^6 \equiv 1 \pmod{7}$, $31 = 6(5) + 1 \Rightarrow 3^{31} \equiv 3^1 = \boxed{3} \pmod{7}$. $2^6 \equiv 1 \pmod{7}$, $35 = 6(5) + 5 \Rightarrow 2^{35} \equiv 2^5 = 32 \equiv \boxed{4} \pmod{7}$. $128 \equiv 9 \pmod{17}$, $9^{16} \equiv 1 \pmod{17}$, $129 = 16(8) + 1 \Rightarrow 128^{129} \equiv 9^1 = \boxed{9} \pmod{17}$.

E. Euler's Theorem & Totient Function

$$a^{\phi(n)} \equiv 1 \pmod{n}, \quad \text{where } \gcd(a, n) = 1, \quad \phi(n) = n \prod_{p \mid n} \left(1 - \frac{1}{p}\right)$$

Steps: (1) factor n , compute $\phi(n)$; (2) write the exponent as a multiple of $\phi(n)$ plus a remainder; (3) reduce using $a^{\phi(n)} \equiv 1$.

Worked example (Lc#6): $7^{402} \pmod{1000}$ $\phi(1000) = 1000(1 - \frac{1}{2})(1 - \frac{1}{5}) = 400$. $402 = 400 + 2 \Rightarrow 7^{402} \equiv 7^{400} \cdot 7^2 \equiv 1 \cdot 49 = \boxed{49} \pmod{1000}$.

Special case — $\gcd(a, n) \neq 1$: Euler's theorem does NOT apply directly; instead look for a short repeating cycle of $a^k \pmod{n}$ by direct computation (see 2022 Q3 below, $5^{1000} \pmod{10}$).

F. Chinese Remainder Theorem (CRT) — NOT in any 2026 Lc material

Statement: If $m_1, \dots, m_k$ are pairwise coprime, the system $x \equiv a_i \pmod{m_i}$ has a unique solution modulo $M = \prod m_i$.

Fast shortcut when all a_i are “the same offset” from m_i :

- If $a_i \equiv -1 \pmod{m_i}$ for every i (i.e. $a_i = m_i - 1$), then $x \equiv -1 \equiv M - 1 \pmod{M}$.
- If $a_i \equiv c \pmod{m_i}$ for the *same* constant c in every congruence, then $x \equiv c \pmod{M}$.

General method (when shortcuts don’t apply):

1. Compute $M = \prod m_i$ and $M_i = M/m_i$.
2. Find $y_i = M_i^{-1} \pmod{m_i}$ (Extended Euclidean, pattern B).
3. $x \equiv \sum_i a_i M_i y_i \pmod{M}$.

Worked example (2023 Q4(a)): $x \equiv 2 \pmod{3}$, $x \equiv 4 \pmod{5}$, $x \equiv 6 \pmod{7}$ Notice $2 \equiv -1 \pmod{3}$, $4 \equiv -1 \pmod{5}$, $6 \equiv -1 \pmod{7}$ — the shortcut applies! $M = 3 \times 5 \times 7 = 105 \Rightarrow x \equiv -1 \equiv \boxed{104} \pmod{105}$.

Worked example (2020 Q2): $x \equiv 1 \pmod{3, 4, 5, 7}$ Same constant $c = 1$ in every congruence $\Rightarrow x \equiv \boxed{1} \pmod{420}$ (where $M = 3 \times 4 \times 5 \times 7 = 420$).

G. Miller-Rabin Primality Test — NOT in any 2026 Lc material

Goal: probabilistically decide whether n is prime.

Algorithm:

1. Write $n - 1 = 2^k \cdot q$ with q odd.
2. Pick a random base a with $1 < a < n - 1$.
3. Compute $x = a^q \pmod{n}$.
4. If $x \equiv 1$ or $x \equiv n - 1 \pmod{n}$: n **passes** this round (probably prime).
5. Otherwise, square $x \leftarrow x^2 \pmod{n}$ up to $k - 1$ more times. If $x \equiv n - 1$ at any point: **passes**.
6. If x never becomes $n - 1$: n is **definitely composite** (base a is a “witness”).

Worked example: test $n = 21$ with base $a = 2$ $n - 1 = 20 = 2^2 \times 5 \Rightarrow k = 2$, $q = 5$.

$$x = a^q \pmod{n} = 2^5 \pmod{21} = 32 \pmod{21} = 11$$

$11 \neq 1$ and $11 \neq 20$. Square once ($k - 1 = 1$ time): $x = 11^2 \pmod{21} = 121 \pmod{21} = 16$. $16 \neq 20$. No more squarings left $\Rightarrow$ **21 is composite** (witness $a = 2$). Indeed $21 = 3 \times 7$. ✓

H. Discrete Logarithm & Cyclic Groups — NOT in any 2026 Lc material

Cyclic group: the nonzero residues mod a prime p , $\{1, 2, \dots, p - 1\}$, form a group under multiplication of order $p - 1$. A **generator (primitive root)** g produces every element as a power $g^1, g^2, \dots, g^{p-1}$ — all distinct.

Discrete log problem: given $g^x \equiv h \pmod{p}$, find x . Easy to compute g^x forward; hard to invert for large p (basis of Diffie-Hellman/ElGamal security). For

small p , just compute powers of g one by one until you hit h .

Worked example (2022 Q3): solve $3^x \equiv 11 \pmod{17}$

$$3^1 = 3, 3^2 = 9, 3^3 = 10, 3^4 = 13, 3^5 = 5, 3^6 = 15, 3^7 = \boxed{11} \pmod{17}$$

$\Rightarrow x = 7$ (general $x = 7 + 16k$, since $3^{16} \equiv 1 \pmod{17}$ — 3 is a primitive root mod 17).

Worked example (2020 Q2): distinct remainders of $b^x \pmod{7}$ for $b = 3$, $x = 1..6$

$$3^1 = 3, 3^2 = 2, 3^3 = 6, 3^4 = 4, 3^5 = 5, 3^6 = 1 \pmod{7}$$

All six remainders $\{3, 2, 6, 4, 5, 1\}$ are **distinct** $\Rightarrow 3$ is a **primitive root mod 7** (cyclic group of order $\phi(7) = 6$ is fully generated by 3).

2024 — Q1(b,c): Fermat's Little Theorem + Euler's Theorem Pair (2.5 + 2.5)

Note on exact numbers

The exact numeric values for 2024 Q1(b,c) were not captured in `_TopicQuestionMap.md`. The question is confirmed to be a **Fermat's Little Theorem + Euler's Theorem pair**, in the same style as every other year. Apply Cheat Sheet patterns D and E directly to whatever a, p, n are given — the method below (using the 2023/2021/2020 seeds) is representative of the expected solution shape.

Question — 2024 Q1(b,c) (representative form)

- (b) Using Fermat's Little Theorem, compute $a^{p-1+r} \bmod p$ for given a, p . [2.5]
 (c) Using Euler's Theorem, find $m \in [0, 9]$ such that $m \equiv a^{1000} \bmod n$. [2.5]

Answer — General Method

(b) **Fermat (pattern D):** Reduce the exponent mod $(p - 1)$, since $a^{p-1} \equiv 1 \pmod{p}$. Then compute the small remaining power directly. (See worked examples: $3^{302} \bmod 11 = 9$, $3^{301} \bmod 11 = 3$, $2^{83} \bmod 5 = 3$, $3^{202} \bmod 11 = 9$ below.)

(c) **Euler (pattern E):** If $\gcd(a, n) = 1$, reduce the exponent mod $\phi(n)$, since $a^{\phi(n)} \equiv 1 \pmod{n}$. If $\gcd(a, n) \neq 1$, find the short repeating cycle directly. (See worked examples: $7^{1000} \bmod 11 = 1$, $7^{1000} \bmod 10 = 1$, $5^{1000} \bmod 10 = 5$ below.)

**2023 — Q1(b,c): Fermat + Euler (2.5+2.5); Q4(a): CRT (5);
Section B Q7(b): Linear Congruence (3)**

Question — 2023 Q1(b,c)

- (b) Using Fermat's Little Theorem, find $3^{302} \bmod 11$. [2.5]
 (c) Using Euler's Theorem, find $m \in [0, 9]$ such that $m \equiv 7^{1000} \bmod 11$. [2.5]

Part (b): Fermat's Little Theorem $3^{302} \bmod 11$

Answer — 2023 Q1(b)

11 is prime, $\gcd(3, 11) = 1 \Rightarrow 3^{10} \equiv 1 \pmod{11}$ (Cheat Sheet D).

$$302 = 10(30) + 2 \Rightarrow 3^{302} \equiv (3^{10})^{30} \cdot 3^2 \equiv 1 \cdot 9 = \boxed{9} \pmod{11}$$

Part (c): Euler's Theorem — $m \equiv 7^{1000} \bmod 11$

Answer — 2023 Q1(c)

$\gcd(7, 11) = 1$, 11 prime $\Rightarrow \phi(11) = 10$ and $7^{10} \equiv 1 \pmod{11}$ (Cheat Sheet E).

$$1000 = 10(100) + 0 \Rightarrow 7^{1000} \equiv (7^{10})^{100} \equiv 1^{100} = \boxed{1} \pmod{11}$$

$m = 1$, which lies in $[0, 9]$. ✓

Question — 2023 Q4(a)

State the Chinese Remainder Theorem and use it to solve: $X \equiv 2 \pmod{3}$, $X \equiv 4 \pmod{5}$, $X \equiv 6 \pmod{7}$. [5]

Answer — 2023 Q4(a)

CRT statement (Cheat Sheet F): If $m_1, \dots, m_k$ are pairwise coprime, the system of congruences $X \equiv a_i \pmod{m_i}$ has a **unique solution modulo** $M = \prod m_i$.

Application: 3, 5, 7 are pairwise coprime, $M = 3 \times 5 \times 7 = 105$.

Observe: $2 \equiv -1 \pmod{3}$, $4 \equiv -1 \pmod{5}$, $6 \equiv -1 \pmod{7}$ — i.e. $X \equiv -1$ in every modulus.

$$\Rightarrow X \equiv -1 \equiv 105 - 1 = \boxed{104} \pmod{105}$$

Check: $104 = 34(3) + 2 \checkmark$, $104 = 20(5) + 4 \checkmark$, $104 = 14(7) + 6 \checkmark$.

General solution: $X = 104 + 105k$.

Question — 2023 Section B Q7(b)

Solve the linear congruence $33X \equiv 18 \pmod{280}$.

[3]

Answer — 2023 Section B Q7(b)

Using Cheat Sheet C: $\gcd(33, 280) = 1$, and Extended Euclidean gives $33^{-1} \equiv 17 \pmod{280}$ (since $1 = 17(33) - 2(280)$).

$$X \equiv 17 \times 18 = 306 \equiv 306 - 280 = \boxed{26} \pmod{280}$$

Check: $33 \times 26 = 858 = 3(280) + 18$. ✓

General solution: $X = 26 + 280k$.

2022 — Q3: Miller-Rabin (3); Discrete Log (3); Fermat (2); Euler (1)

Question — 2022 Q3

- (a) State the Miller-Rabin primality algorithm and apply it to test a given number. [3]
- (b) Given $11 \equiv 3^X \pmod{17}$, explain how to find X using cyclic groups. [3]
- (c) Using Fermat's Little Theorem, find $2^{83} \bmod 5$. [2]
- (d) Using Euler's Theorem, find $X \in [0, 9]$ such that $X \equiv 5^{1000} \bmod 10$. [1]

Part (a): Miller-Rabin Primality Test

Answer — 2022 Q3(a)

Algorithm (Cheat Sheet G):

1. Write $n - 1 = 2^k q$ with q odd.
2. Pick random base a , $1 < a < n - 1$. Compute $x = a^q \bmod n$.
3. If $x \equiv 1$ or $x \equiv n - 1$: “probably prime”.
4. Else square x up to $k - 1$ times; if it ever equals $n - 1$: “probably prime”.
5. If never $\equiv n - 1$: **definitely composite**.

Applied example: $n = 21$, $a = 2$. $n - 1 = 20 = 2^2 \times 5 \Rightarrow k = 2, q = 5$.
 $x = 2^5 \bmod 21 = 32 \bmod 21 = 11$. $11 \neq 1, 20$. Square: $x = 11^2 \bmod 21 = 121 \bmod 21 = 16 \neq 20$. No squarings left $\Rightarrow$ **21 is composite** (matches $21 = 3 \times 7$).

Part (b): Discrete Log via Cyclic Groups — $3^X \equiv 11 \pmod{17}$

Answer — 2022 Q3(b)

Cyclic group concept (Cheat Sheet H): $\{1, \dots, 16\}$ under multiplication mod 17 is a cyclic group of order 16. If 3 is a generator (primitive root), its powers $3^1, \dots, 3^{16}$ run through all 16 nonzero residues exactly once. Find X by computing successive powers of 3 until reaching 11:

$$3^1 = 3, 3^2 = 9, 3^3 = 10, 3^4 = 13, 3^5 = 5, 3^6 = 15, 3^7 = \boxed{11} \pmod{17}$$

$\Rightarrow X = 7$ (general $X = 7 + 16k$, since $3^{16} \equiv 1 \pmod{17}$).

Part (c): Fermat's Little Theorem — $2^{83} \bmod 5$

Answer — 2022 Q3(c)

5 is prime, $\gcd(2, 5) = 1 \Rightarrow 2^4 \equiv 1 \pmod{5}$.

$$83 = 4(20) + 3 \Rightarrow 2^{83} \equiv (2^4)^{20} \cdot 2^3 \equiv 1 \cdot 8 \equiv \boxed{3} \pmod{5}$$

Part (d): Euler's Theorem (special case) — $X \equiv 5^{1000} \bmod 10$

Answer — 2022 Q3(d)

$\gcd(5, 10) = 5 \neq 1$, so Euler's theorem does **not** apply directly (Cheat Sheet E special case). Instead compute the pattern directly:

$$5^1 = 5, \quad 5^2 = 25 \equiv 5 \pmod{10}, \quad 5^3 \equiv 5 \pmod{10}, \dots$$

$$5^n \equiv 5 \pmod{10} \text{ for every } n \geq 1 \Rightarrow X = \boxed{5}.$$

2021 — A3: Discrete Log (2.75); Ext. Euclidean Mult. Inverse (2); Fermat (2); Euler (2)

Question — 2021 A3

- (a) Discrete logarithm / cyclic group problem. [2.75]
- (b) Find the multiplicative inverse of 1234 mod 4321 using the Extended Euclidean Algorithm. [2]
- (c) Using Fermat's Little Theorem, find $3^{301} \bmod 11$. [2]
- (d) Using Euler's Theorem, find $X \in [0, 28]$ such that $X^{45} \equiv 6 \pmod{35}$. [2]

Part (a): Discrete Log / Cyclic Group

Answer — 2021 A3(a)

Exact numbers not captured in `_TopicQuestionMap.md`. Apply Cheat Sheet H: identify the cyclic group $\{1, \dots, p-1\}$ under multiplication mod p , confirm/use a generator (primitive root), and compute successive powers of the base until the target residue is reached — same method as 2022 Q3(b) and 2020 Q2 above.

Part (b): Multiplicative Inverse of 1234 mod 4321

Answer — 2021 A3(b)

Using Cheat Sheet B, the Euclidean steps are:

$$4321 = 3(1234) + 619, \quad 1234 = 1(619) + 615, \quad 619 = 1(615) + 4, \quad 615 = 153(4) + 3, \quad 4 = 1(3) + 1$$

Back-substituting gives $1 = 309(4321) - 1082(1234)$, so

$$1234^{-1} \equiv -1082 \equiv 4321 - 1082 = \boxed{3239} \pmod{4321}$$

Check: $1234 \times 3239 = 3,996,926 = 925(4321) + 1$. ✓

Part (c): Fermat's Little Theorem — $3^{301} \bmod 11$

Answer — 2021 A3(c)

$$3^{10} \equiv 1 \pmod{11}. \quad 301 = 10(30) + 1 \Rightarrow 3^{301} \equiv (3^{10})^{30} \cdot 3^1 \equiv 1 \cdot 3 = \boxed{3} \pmod{11}.$$

Part (d): Euler's Theorem — $X^{45} \equiv 6 \pmod{35}$, $X \in [0, 28]$

Answer — 2021 A3(d)

$35 = 5 \times 7$. Solve via CRT (Cheat Sheet F) by working mod 5 and mod 7 separately.

Mod 5: $\phi(5) = 4$, $45 = 4(11) + 1 \Rightarrow X^{45} \equiv X^1 \pmod{5}$ (for $\gcd(X, 5) = 1$). Need $X \equiv 6 \equiv 1 \pmod{5}$.

Mod 7: $\phi(7) = 6$, $45 = 6(7) + 3 \Rightarrow X^{45} \equiv X^3 \pmod{7}$. Need $X^3 \equiv 6 \pmod{7}$. Testing $X = 1..6$: $3^3 = 27 \equiv 6$, $5^3 = 125 \equiv 6$, $6^3 = 216 \equiv 6 \pmod{7}$ — so $X \equiv 3, 5, 6 \pmod{7}$.

Combine with CRT: need $X \equiv 1 \pmod{5}$ AND $X \in \{3, 5, 6\} \pmod{7}$, with $X \in [0, 28]$. Values $\equiv 1 \pmod{5}$ in range: 1, 6, 11, 16, 21, 26. Checking mod 7: $6 \pmod{7} = 6 \checkmark$, $26 \pmod{7} = 5 \checkmark$ (others fail).

$$\Rightarrow X = \boxed{6} \text{ or } X = \boxed{26}$$

Sanity check for $X = 6$: $6^2 = 36 \equiv 1 \pmod{35}$, so $6^{45} = 6^{2(22)+1} \equiv 1^{22} \cdot 6 = 6 \pmod{35}$. $\checkmark$

**2020 — Q2: CRT (4.75) + Discrete Log/Cyclic Groups (2+2);
Q3: Ext. Euclidean (4.75) + Fermat & Euler (2+2)**

Question — 2020 Q2

- (a) Use CRT to solve $X \equiv 1 \pmod{3}$, $X \equiv 1 \pmod{4}$, $X \equiv 1 \pmod{5}$, $X \equiv 1 \pmod{7}$. [4.75]
- (b) Discrete logarithm / cyclic group concept. [2]
- (c) List the distinct remainders of $b^x \pmod{7}$ for $x = 1, \dots, 6$. [2]

Part (a): CRT — $X \equiv 1$ modulo 3, 4, 5, 7

Answer — 2020 Q2(a)

3, 4, 5, 7 are pairwise coprime, $M = 3 \times 4 \times 5 \times 7 = 420$ (Cheat Sheet F). Since $X \equiv 1$ for **every** modulus (same constant $c = 1$):

$$X \equiv \boxed{1} \pmod{420}$$

General solution: $X = 1 + 420k$ (smallest positive: $X = 1$, or $X = 421$ if a nonzero/positive-only answer is expected).

Part (b): Discrete Log / Cyclic Group Concept

Answer — 2020 Q2(b)

Exact numbers not captured in `_TopicQuestionMap.md`. Use Cheat Sheet H: explain that $\{1, \dots, p-1\}$ under multiplication mod p is a cyclic group of order $p-1$; a generator (primitive root) g produces all elements via $g^1, \dots, g^{p-1}$; the discrete log problem is to invert $g^x \equiv h \pmod{p}$ for x — easy forward, hard backward for large p .

Part (c): Distinct Remainders of $b^x \pmod{7}$, $x = 1..6$

Answer — 2020 Q2(c)

Take $b = 3$ (a primitive root mod 7):

$$3^1 = 3, 3^2 = 2, 3^3 = 6, 3^4 = 4, 3^5 = 5, 3^6 = 1 \pmod{7}$$

The remainders $\{3, 2, 6, 4, 5, 1\}$ are **all distinct** $\Rightarrow 3$ generates the full cyclic group of order $\phi(7) = 6$, i.e. 3 is a **primitive root mod 7**.

Question — 2020 Q3

- (a) Find the multiplicative inverse of 550 mod 1769 using the Extended Euclidean Algorithm. [4.75]
- (b) Using Fermat's Little Theorem, find $3^{202} \bmod 11$. [2]
- (c) Using Euler's Theorem, find $7^{1000} \bmod 10$. [2]

Part (a): Multiplicative Inverse of 550 mod 1769

Answer — 2020 Q3(a)

Using Cheat Sheet B, the Euclidean steps on (1769, 550) reduce (after full back-substitution) to:

$$\begin{aligned} 1 &= 550(550) - 171(1769) \\ \Rightarrow 550^{-1} &\equiv \boxed{550} \pmod{1769} \end{aligned}$$

Check: $550 \times 550 = 302,500 = 171(1769) + 1$. ✓ (550 happens to be its own inverse mod 1769.)

Part (b): Fermat's Little Theorem — $3^{202} \bmod 11$

Answer — 2020 Q3(b)

$$3^{10} \equiv 1 \pmod{11}. \quad 202 = 10(20) + 2 \Rightarrow 3^{202} \equiv (3^{10})^{20} \cdot 3^2 \equiv 1 \cdot 9 = \boxed{9} \pmod{11}.$$

Part (c): Euler's Theorem — $7^{1000} \bmod 10$

Answer — 2020 Q3(c)

$$\gcd(7, 10) = 1, \quad \phi(10) = 10\left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{5}\right) = 4 \Rightarrow 7^4 \equiv 1 \pmod{10}.$$

$$1000 = 4(250) + 0 \Rightarrow 7^{1000} \equiv (7^4)^{250} \equiv 1^{250} = \boxed{1} \pmod{10}$$

Exam Strategy Summary

High-Yield Checklist for Number Theory & Modular Arithmetic (5/5 years, 5–9 marks)

1. **Extended Euclidean** → **multiplicative inverse** (Cheat Sheet B): asked 2020, 2021 — 4.75/2 marks. Practice the back-substitution table until it's mechanical.
2. **Fermat's Little Theorem** $a^{p-1} \equiv 1 \pmod{p}$ (Cheat Sheet D): asked EVERY year 2020–2024, recurring seed $3^{20x} \pmod{11}$. Reduce exponent mod $(p-1)$.
3. **Euler's Theorem** $a^{\phi(n)} \equiv 1 \pmod{n}$ (Cheat Sheet E): asked EVERY year 2020–2024, recurring seed $7^{1000} \pmod{(10 \text{ or } 11)}$. Watch for the $\gcd(a, n) \neq 1$ special case (2022 Q3(d)).
4. **CRT** (Cheat Sheet F, NO-SOURCE): asked 2020, 2023 — look for the “ $\equiv -1$ ” or “same constant” shortcuts first before doing full CRT arithmetic.
5. **Miller-Rabin** (Cheat Sheet G, NO-SOURCE): asked 2022 — memorize the 6-step algorithm + the $n = 21, a = 2$ worked example.
6. **Discrete log / cyclic groups** (Cheat Sheet H, NO-SOURCE): asked 2020–2022 — compute successive powers of the base by hand until the target is reached; know the term “primitive root”.
7. **Linear congruence** $ax \equiv b \pmod{m}$ (Cheat Sheet C): asked 2023 — find a^{-1} via Extended Euclidean, multiply by b .

This topic feeds directly into RSA (Topic 8) — the Extended Euclidean / multiplicative inverse skill (B) and Euler's totient (E) are the exact mechanics behind RSA key generation. Master this topic first.